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An experimental and kinetic modeling study of *n*-dodecane pyrolysis and oxidation

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ABSTRACT

The current study investigates *n*-dodecane (n-$C_{12}H_{26}$) pyrolysis and oxidation kinetics in the temperature regime of 1000–1300 K in the Stanford Variable Pressure Flow Reactor facility. The reactor environment is vitiated and the experiments were conducted at atmospheric pressure. Species time history data were collected for *n*-dodecane and oxygen, as well as for 12 intermediate and product species over a span of 1–40 ms residence times using real time gas chromatography. The experimental data were compared against the predictions of four detailed kinetic models. The results showed that the fuel oxidation proceeds through an early pyrolytic stage where the fuel breaks down into smaller hydrocarbon fragments, including mostly C_{2-4} alkenes, and a late oxidation stage where the fragments oxidize to CO. The kinetic models were observed to diverge notably in their predictions from one another. Sensitivity and flux analysis identified the cause of the divergences to differences in the small hydrocarbon chemistry modeling. Finally, the flow reactor data were used to demonstrate how model uncertainty minimization can improve model predictions. It is shown that after uncertainty minimization against a selected set of *n*-dodecane combustion data, the predictions of the resulting optimized model are improved notably for all existing *n*-dodecane data sets tested, including those of the current flow reactor study that were not part of the optimization target list.

1. Introduction

The development of the next generation of efficient, fuel flexible gas turbine propulsion systems requires the implementation of large eddy simulation (LES) based turbulent combustion models that incorporate a combustion chemistry model for the aviation fuels [1]. Such needs have motivated a significant effort toward the construction of detailed and reduced kinetic models for kerosene fuels. Typical jet fuels (JP-8 or Jet-A) are variable blends of thousands of hydrocarbon compounds [2,3]. While the nominal functional group composition of aviation fuels are found to be approximately 20% *n*-alkanes, 40% *iso*-alkanes, 20% cyclo-alkanes and 20% aromatics, variations are observed among different batches of jet fuels [4]. A detailed modeling approach based on brute force reconstruction of the entire composition of the fuel molecules and their reaction intermediates is therefore impossible. Work has focused instead on identifying and studying a few hydrocarbon species which, when blended together in certain proportions, create fuel surrogates that mimic important chemical and physical properties of an aviation fuel of interest. *n*-Dodecane (n-$C_{12}H_{26}$) has been considered as a component species in most surrogate blends for jet fuels [3, 5-8]. *n*-Dodecane has also been proposed as a component of surrogates that mimic RP-1, the kerosene fuel used in rocket engines [9]. Several detailed and semi-detailed kinetic models have been proposed for heavy *n*-alkanes including *n*-dodecane. The combustion chemistry group at the Lawrence Livermore National Laboratory (LLNL) has developed detailed kinetic models for *n*-alkanes up to *n*-hexadecane [10] that consider both high and low temperature chemistry. The Centre national de la recherche scientifique (CNRS) group in Orleans has extended their hierarchical model to include *n*-dodecane chemistry [11]. An automated kinetic scheme generator software, EXGAS, was used to generate *n*-alkane models by the Nancy group [12]. Another kinetic model generator, MAMOX ++, developed by the CRECK group in Milano has been used to advance a semi-detailed model for *n*-alkane oxidation [13]. The Jet Surrogate Fuel (JetSurF) model [14] includes the chemistry for the oxidation of *n*-alkanes up to *n*-dodecane and lumped low temperature chemistry sub-model for *n*-alkane oxidation in the negative temperature coefficient (NTC) region. An elucidation of the chemical kinetic behaviors of *n*-dodecane in fundamental experimental apparatus (e.g., ignition delay in shock tubes, flame speed and

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extinction strain rates, species profiles in stirred and flow reactors and shock tubes) is essential for the continued improvements and test of these kinetic models.

Flow reactors and jet stirred reactors (JSRs) have been used to obtain intermediate and product profiles during *n*-dodecane pyrolysis and oxidation between 500 and 1050 K. Dahm et al. [15] investigated the thermal decomposition of 0.336% *n*-dodecane diluted in N_2 in a stainless steel isothermal plug flow reactor between 950 and 1050 K at 1 atm pressure. Herbinet et al. [16] studied the pyrolysis of 2% *n*-dodecane in He from 773 to 1073 K at 1 atm using a quartz JSR. Observations made in both studies were used to test a detailed kinetic model generated by EXGAS. Kurman et al. [17] studied lean oxidation of *n*-dodecane in the low temperature and the NTC regimes (550–830 K) and at high pressure (~8 atm) in a pressurized turbulent flow reactor. The time profiles of species were compared with predictions from kinetic models. Using a JSR, Mze-Ahmed et al. [11] studied the oxidation kinetics of *n*-dodecane over a range of equivalence ratios for temperatures from 550 to 1150 K and at an elevated pressure (10 atm). The species concentrations were compared with the predictions of the CNRS model. Veloo et al. [18] compared the relative reactivity of $C_{7\text{-}14}$ *n*-alkanes at stoichiometric conditions between 500 and 1000 K and at 8 atm in the Princeton Turbulent Flow Reactor facility. The low temperature reactivity region was found to be between 630 and 700 K at that pressure. While the high temperature reactivities were found to be similar among the light and heavy *n*-alkanes tested, their reaction behaviors diverged in the NTC region. Shock tube experiments have been conducted to study ignition delay times of *n*-dodecane and to investigate the species time histories. Vasu et al. [19] determined the ignition delay and the OH time histories of *n*-dodecane-air mixtures in the Stanford High Pressure Shock Tube facility for temperatures from 727 to 1422 K and pressures from 15 to 34 atm at 0.5 and 1.0 equivalence ratio. The NTC behavior was observed to be between 750 and 950 K in that pressure range. Shen et al. [20] measured the ignition delay of *n*-alkanes from *n*-heptane to *n*-hexadecane between 786 and 1396 K and 9–58 atm in a heated shock tube under lean and stoichiometric conditions. The results were found to be consistent with data reported by Vasu et al. [19]. Comparisons of the data indicated that the ignition delay times are well within the experimental uncertainties for all the reactant fuels tested. MacDonald et al. [21] investigated the thermal decomposition of *n*-dodecane in a shock tube for temperature and pressure ranges of 1138-1522 K and 17-23 atm and found ethylene to be the dominant pyrolysis product. Species time history profiles were also determined for *n*-dodecane, CO_2, OH and C_2H_4 by Davidson et al. [22] at stoichiometric conditions between 1300 and 1600 K temperature, 2 atm pressure. The fuel decomposition was found to be rapid and decoupled in time from the oxidation of decomposed products. Comparison of the data with predictions from the JetSurF and the LLNL model showed that while the species yields were well predicted, the match for the species time history profiles was not as good. Malewicki and Brezinsky [23] used a high-pressure single-pulse shock tube to investigate the kinetics of intermediate and product species formation for *n*-decane and *n*-dodecane pyrolysis and oxidation at 850–1750 K temperature, 19–74 atm pressures, and also found that the fuel decomposition was largely decoupled from the oxidation of the decomposition products.

The current work is a comprehensive study of intermediate to high temperature *n*-dodecane oxidation with several related objectives. Prior studies have focused mostly on the low temperature (500 K < T < 900 K) or the high temperature (1300 K < T < 1700 K) regions of *n*-dodecane pyrolysis and oxidation. The intermediate temperature window of 1000–1300 K has been relatively less studied. The upper end of this temperature range corresponds to the preheat zone of flames where heavier fuel molecules are expected to decompose into smaller hydrocarbon fragments [24]. Therefore, the intermediate temperature pyrolysis and oxidation kinetics is expected to play a critical role in the overall combustion process. Thus, the current work investigates the pyrolysis and oxidation kinetics of *n*-dodecane in the intermediate temperature regime using the Stanford Variable Pressure Flow Reactor (VPFR) facility. The species time profiles are obtained under four separate conditions, covering pyrolysis and oxidation, the reactant stoichiometry, and temperature, all at 1 atm pressure. These data are used to assess the accuracy of several kinetic models available in the literature [10-13] in predicting *n*-dodecane oxidation in the intermediate temperature regime. As a tertiary objective, one of the four models (the JetSurF model) was subject to systematic optimization and uncertainty minimization with the goal to demonstrate how the model uncertainty may be reduced. The optimization uses the Method of Uncertainty Minimization by Polynomial Chaos Expansion (MUM-PCE) [25-28] against a target set of *n*-dodecane combustion data, notably the shock tube multi-species time history data of *n*-dodecane oxidation of Davidson et al. [22]. The current flow reactor data were used in conjunction with other data sets from the literature to test the optimized JetSurF model with respect to improvements resulting from the uncertainty minimization analysis. This element of the work also facilitates an assessment of the consistency of the literature experimental data on *n*-dodecane pyrolysis and oxidation. Lastly, we examine the causes for the discrepancies observed among the different reaction models using sensitivity and flux analysis and make appropriate recommendations for future work needed in order to make improvements of the existing reaction models.

2. Experimental method

2.1. The flow reactor facility

The current study uses the Stanford VPFR facility to investigate the pyrolysis and oxidation kinetics of *n*-dodecane in the temperature regime between 1000 and 1300 K at 1 atm pressure. The flow reactor, described elsewhere [29-31], is comprised of a vertical quartz tube connected to a converging–diverging duct into which a high momentum cross-flow jet of vaporized fuel/N_2 mixture is injected upstream of the reaction zone (Fig. 1). The combustion products of a H_2/air flame stabilized on a water-cooled McKenna burner [32] at the base of the reactor provide a hot vitiated flow in which the injected fuel undergoes reaction under near adiabatic conditions. Three electrical resistance heaters surround the reactor tube and compensate for any heat loss from the reaction zone. There is an initial stage of cross-flow injection of pure nitrogen that introduces turbulence and decreases the temperature of the vitiated products. The quartz tube is enclosed in a pressure vessel designed to withstand pressures up to 50 atm. Gas flows into the reactor are metered through PID controlled critical flow orifices. The *n*-dodecane vaporization system consists of a stainless steel cylinder partially filled with liquid *n*-dodecane that is kept immersed in a heated thermal bath. A preheated stream of N_2 gas is introduced into the cylinder by using a commercial stainless steel sparger with a porous area of 8.36 cm^2. The sparging process enhances gas–liquid contact allowing the N_2 carrier gas bubbles to pick up vaporized *n*-dodecane as they rise through the fuel tank and enter heated SS tubing at the vaporizer exit. Downstream of the vaporizer, a flame ionization detector is used to measure the fuel concentration in the carrier gas and to determine if the vaporizer stream has attained steady state. The temperature and species concentration along the reactor axis are measured using a thermocouple and a species extraction probe that are attached to a translation stage. The Pt/13%Rh–Pt thermocouple bead is coated with a passivating SiO_2 film, while the propylene glycol cooled extraction probe has a choked inlet to aerodynamically quench the extracted gas and “freeze” the species mixture composition at the

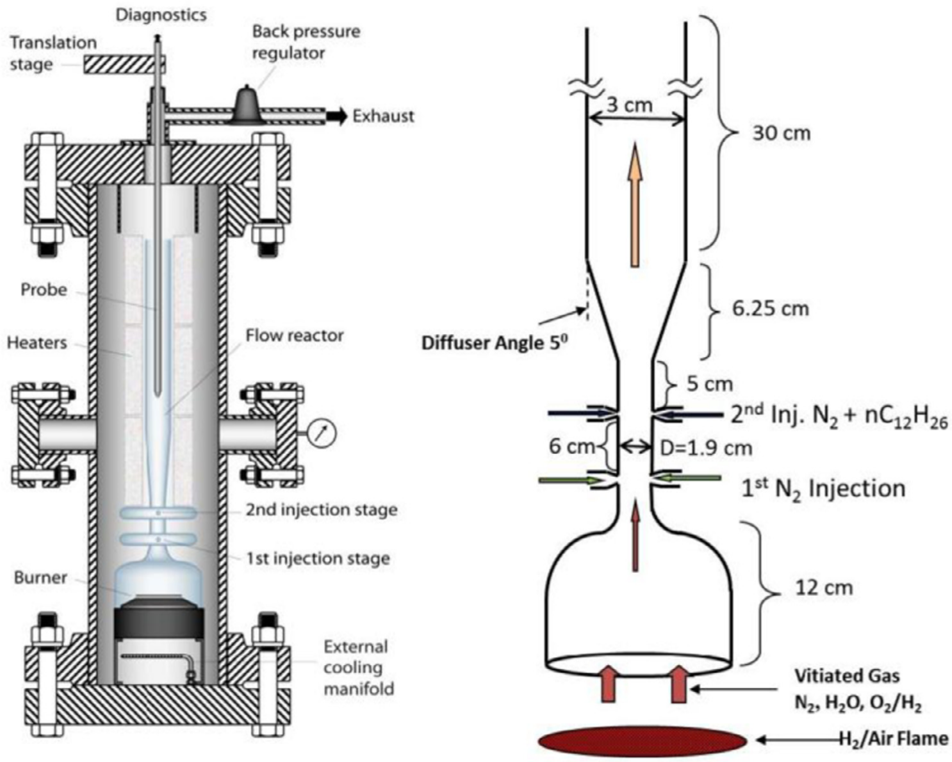

Fig. 1. The VPFR facility and a schematic of the quartz flow reactor.

extraction point. A torpedo laser level mounted on a vertical alignment frame is used to ensure that the burner assembly, the quartz flow reactor, the pressure vessel and the probe mounts are assembled co-axially and are in a vertical position with respect to the steel platform base. This ensures that the probe tip does not deviate from the reactor centerline during translation. The extracted sample gas is sent to a 4 column micro gas chromatograph (micro GC 3000 by Inficon) that provides real time detection of the concentration levels of the stable compounds (H_2,O_2, CO, CO_2, CH_4, CH_2O, C_2H_4,C_2H_2,C_2H_6, C_3H_6, 1-C_4H_8, 1-C_5H_{10}, 1-C_6H_{12}, 1-C_7H_{14} and *n*-$C_{12}H_{26}$) within a matter of minutes. A non-dispersive infrared analyzer (NDIR) and a paramagnetic analyzer (PMA) are also used for near instantaneous measurements of CO, CO_2 and O_2 within the flow reactor. The total uncertainty in species concentration is $\pm$2% to $\pm$5% for most species. The radiation corrected values for the temperature are in the 3 to 1 K range along the length of the reactor due to the presence of the heaters and the hot pressure vessel surrounding the flow reactor, while the net uncertainty in the measured temperature data is $\pm$4 K.

2.2. The flow reactor modeling

A previous two-dimensional finite-volume study of the flow reactor [30,33] demonstrated that a simple one-dimensional plug flow assumption is valid within the reaction zone as long as the species and temperature measurements are conducted along the reactor centerline. The velocity profiles along the reactor axis were measured in an earlier study [29,34] and residence time estimates are made on the basis of a dimensionless mass flow rate model [31] that uses the axial gas velocity, the axial temperature and the reactor diameter as inputs. The uncertainty in the residence time estimates varies from 0.5 to 3 ms in the 0–40 ms residence time window of the experiment. The initial fuel mixing zone has been modeled using a one-dimensional mixing reacting model developed by Zwietering [35,36]. The model implementation has been described and discussed previously in [29,30]. Other techniques for modeling the mixing zone exist [37]. However, the popular time-shifting approach is unsuitable for cases where a significant fraction of the initial reactant is consumed in the mixing zone itself, while the initial conditioning approach described in [37] relies on the fidelity of the chemical model and cannot be used for a comparative analysis among the different reaction models which is a major goal of the current study. The Zwietering approach does not depend on the chemical kinetic predictions to model the mixing region. Instead, its mixing time scale (τ_e) is determined for each experimental condition by injecting a passive scalar (CO_2) through the injector and measuring its rate of dilution into the bath gas. Typical mixing times range from 0.7 to 2.4 ms (Table 1). Figure 2 presents the results of one sample CO_2 mixing experiment where the CO_2 dilution data have been used to fit an exponential mixing curve with the unmixed CO_2 stream concentration serving as the initial starting point of the fit at 0 ms. The mixing reacting model was able to correctly predict the known rate of decomposition of ethyl-chloride into ethylene and HCl within the mixing region of the reactor thereby establishing the robustness of the semi-empirical approach based on the mixing characteristic of a non-reacting scalar. Furthermore, numerical analyses showed little to no change in the predicted species concentration with the mixing time increased or decreased by 30%, indicating that the mixing time and model adopted here is not a significant source of uncertainty in the modeling results downstream of the mixing region. The mixing reacting model was implemented in the CHEMKIN-PRO PFR module with two inlet streams in conjunction with detailed kinetic models to generate predictions for species profiles that can be compared with the measured data.

2.3. Detailed kinetic models

Four detailed kinetic models have been selected for comparing the experimental data with the mixing-reaction PFR model

Table 1
Conditions for *n*-dodecane pyrolysis and oxidation experiments.

Parameter	1000 K	Pyrolysis 1050 K rich oxidation	1220 K rich oxidation	1170 K lean oxidation
Equivalence	∞	1.56	1.32	0.79
Background temperature (K)	1000	1050	1220	1170
Pressure (atm)	1	1	1	1
$C_{12}H_{26}$ (ppmv)	282	501	569	328
Nominal H_2O (%)	22	19.8	25	24.2
Nominal O_2 (%)	0	0.59	0.79	0.77
Nominal H_2 (%)	6.6	0	0	0
N2 (%)	71	79	73.5	74.4
Total residence time (ms)	41.3	37.9	40.3	40.8
Mixing time (ms)	1.4	0.69	1.6	2.4

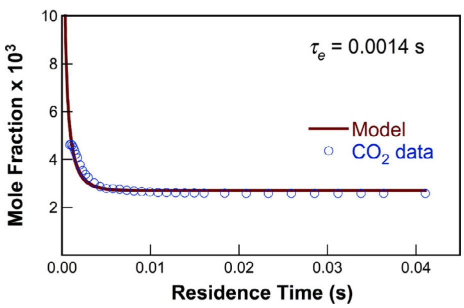

Fig. 2. CO_2 mixing profile shown for the 1000 K pyrolysis experiment at 1 atm.

simulations. JetSurF (version 1.0) [14] is a detailed kinetic model that describes the high temperature combustion behavior of *n*-alkanes up to *n*-dodecane. The model is comprised of 194 species and 1459 reactions and is an extension of the USC-Mech II model [38] that describes the H_2/CO/C_{1-4} combustion chemistry. USC-Mech II contains 111 species and 784 reactions and in JetSurF 1.0, they are appended with 663 additional reactions and 79 species that extend the model to describe high temperature oxidation and pyrolysis chemistry of C_{5-12} *n*-alkanes. The main reaction classes that have been considered include C–C fission, H abstraction reactions from alkanes to create alkyl radicals, mutual isomerization of alkyl radicals by H transfer, β-scission of alkyl radicals, H abstraction from alkenes and decomposition of *n*-alkenyl radicals. Rate parameters for reactions involving higher alkanes, alkyl radicals and alkenes were estimated from analogous reactions of *n*-butane, butyl radicals and 1-butene. JetSurF 1.0 is not intended for the prediction of any low temperature fuel oxidation behavior. However, a 4-species, 12-step lumped low temperature model based on Bikas and Peters [39] has been included to capture the impact of low temperature chemistry on *n*-dodecane oxidation at the intermediate temperature region (900–1200 K).

The combustion chemistry group at the LLNL has developed a reaction model for a wide range of hydrocarbon fuels. Normal and branched chain alkanes have received particular attention. The LLNL *n*-alkane model is based on a detailed kinetic model of *n*-heptane oxidation originally proposed by Curran et al. [40] that was subsequently extended to cover *n*-alkanes up to *n*-hexadecane [10]. The LLNL model used in the current work is the web version of the model released in 2011. Ten reaction classes are identified for the high temperature *n*-alkane oxidation and pyrolysis, while 25 reaction classes are considered for the NTC and the low temperature regimes. The *n*-dodecane model with detailed low temperature chemistry contains 1282 species and 5030 elementary reactions.

The CNRS group at Orleans has also developed detailed chemical kinetic models of normal, branched and cyclo alkanes [41]. Recently, they investigated *n*-dodecane and *n*-undecane oxidation kinetics in a JSR and developed a detailed kinetic model comprising of 1377 species and 5864 reactions that showed good accuracy against *n*-dodecane experimental data [11]. This model (henceforth referred to as the CNRS model) is the third detailed kinetic scheme against which our flow reactor data are compared.

The CRECK (Chemical Reaction Engineering and Chemical Kinetics) modeling group at Milan uses an automated code to generate semi-detailed models for hydrocarbon oxidation and pyrolysis. The kinetic scheme developed by this approach for *n*-alkanes up to *n*-hexadecane is described in [13,42] and is henceforth referred to as the CRECK model. The semi-detailed model uses lumping procedures to reduce the number of species and reaction pathways in an automatically generated fully detailed model while preserving the descriptive capabilities of the detailed scheme. The simplifications include the assumption of quasi-steady state for the various isomers of a given alkyl radical, classification of the reactions into 10 basic classes and the lumping of intermediate radicals and isomers of the primary products into a few groups of pseudo-species. The model consists of 460 species and 16,039 reactions.

3. Experimental data and model comparisons

Four sets of experimental conditions are investigated in the VPFR facility as shown in Table 1. The 1000 K pyrolysis and the 1050 K rich oxidation experiments are designed to capture the early stage chemistry of *n*-dodecane decomposition, while the 1220 K rich and 1170 K lean oxidation experiments are designed to capture the later oxidation and heat generation stages of the reaction. The temperature is measured along the axis of the reactor. In oxidation experiments where heat release may be significant, the temperature is recorded for both reacting and non-reacting (no fuel injection) conditions. Species profiles are recorded using the combined GC/NDIR/PMA real-time diagnostic system and the values are presented along with experimental scatter (vertical error bars) and residence time uncertainty estimates (horizontal error bars). The residence time uncertainties range from 0.5 ms at the early times to 3 ms near the reactor exit. The uncertainties in the species measurements (2–5%) are too small to be visible and are not included in the figures. The species data are compared against the predictions from the four detailed kinetic models (JetSurF 1.0, CNRS, LLNL and CRECK). In all cases, the experimentally determined temperature is used to fit a temperature profile in the reactor and this profile is used as an input.

The vitiated gas contains either hydrogen or oxygen, depending on whether it is a pyrolysis or an oxidation experiment. As an initial condition, the amount of hydrogen or oxygen was accounted for as seen in Table 1. Water is expected to enhance the third-body effect, which has been considered in the modeling effort. Additionally, flame simulations suggest that 10 ppm of H atoms can be present in the vitiated gas for the pyrolysis experiments and were thus included

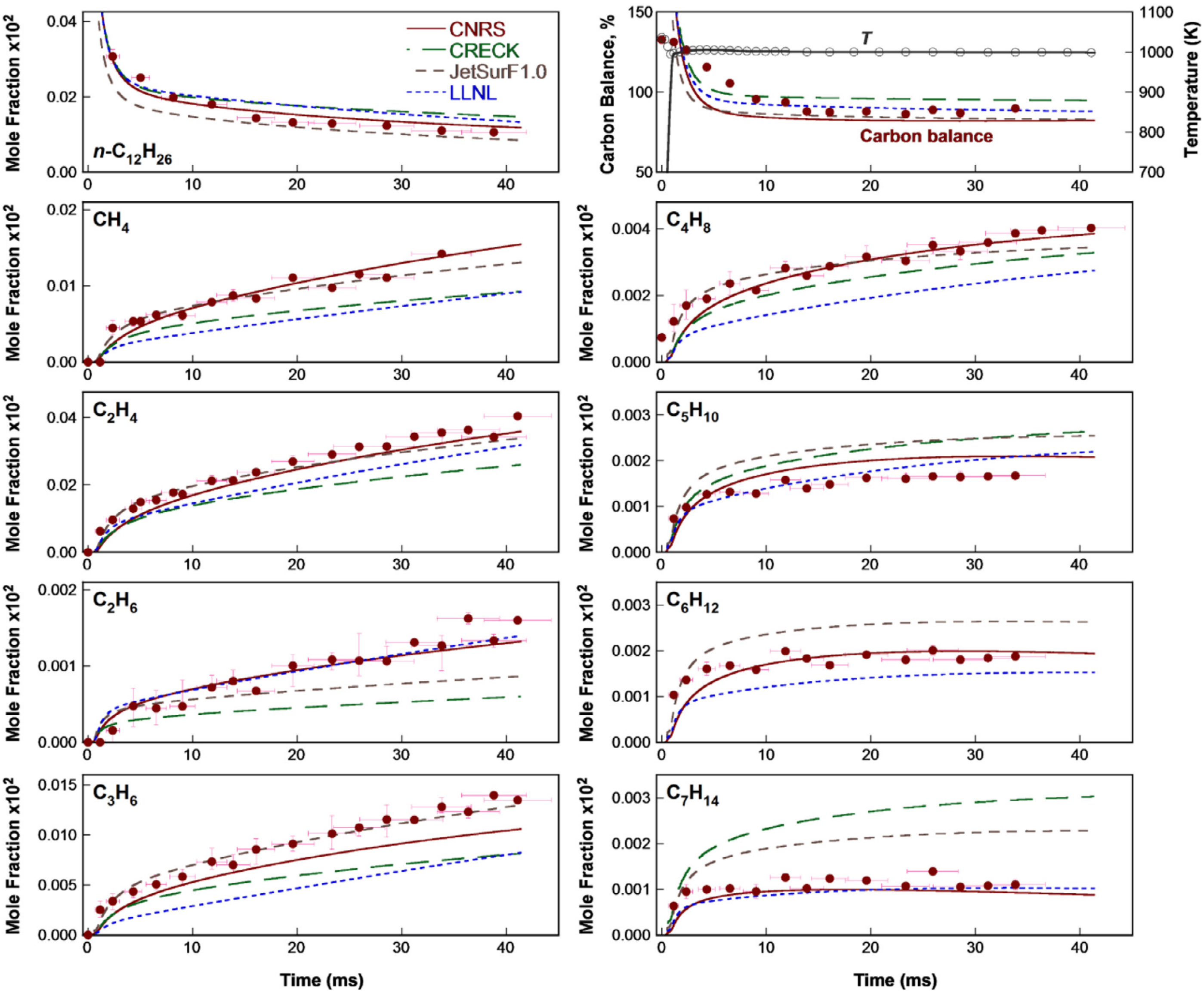

Fig. 3. Species mole fraction, carbon balance and temperature data (symbols) for the 1000 K pyrolysis experiment compared against predictions (lines) from the detailed kinetic models.

in the initial condition of our simulation. For the oxidation conditions, radicals are not present in significant quantities in the vitiated gas.

The data for the pyrolysis experiment at 1000 K temperature and 1 atm pressure are shown in Fig. 3. 282 ppmv of *n*-dodecane (after complete mixing) is injected into the flow reactor and the temperature measurements reflect the initial drop due to the mixing of the colder fuel-nitrogen stream with the hotter vitiated gas. Beyond the initial drop, the axial temperature remains nearly constant at about 1000 K until the reactor exit. A fit to the temperature data, including the estimate that the fuel-N_2 stream is at 400 K at the second injection point, is used as an input for the model. The mixing constant is 1.4 ms, and mixing effects becomes negligible beyond 5–6 ms residence times. Concentration measurements show that around 63% of the *n*-dodecane decomposes by 40 ms with C_2H_4, C_3H_6, CH_4 and 1-C_4H_8 as the pyrolysis products in descending order of importance. Ethane and small amounts of higher alkenes up to 1-heptene are detected as well. The measured species account for 90% of the total fuel carbon in the fully mixed region of the reaction zone. Both the model and the experimental carbon balance is seen to be greater than 100% in the mixing region (0–5 ms) where as expected, the fuel stream has not mixed completely with the bath gas.

The pyrolysis products have not attained steady-state plateau values at the reactor exit as shown in Fig. 3. With this caveat, the exit ethylene yield (defined as the maximum moles of ethylene per initial moles of the fuel) is 1.4. In terms of model predictions, the CNRS model performs best in predicting the experimental profiles, while the JetSurF 1.0 model predicts the major products (C_2H_4, CH_4, C_3H_6 and 1-C_4H_8) as well as the fuel decomposition rate in a satisfactory manner. In contrast, the LLNL and the CRECK models predict a slower rate of fuel decomposition into intermediate product species. The CRECK model predicts no 1-hexene production at all while 1-heptene is predicted to be three times the measured concentrations.

The species concentration data for the rich 1050 K oxidation experiment are presented in Fig. 4. The equivalence ratio is 1.56 and the injected *n*-dodecane mole fraction is 501 ppmv. The experimental conditions capture the early stages of fuel breakdown before oxygen-consuming reactions assume importance. This may be seen from the negligible decrease in the oxygen concentration from the

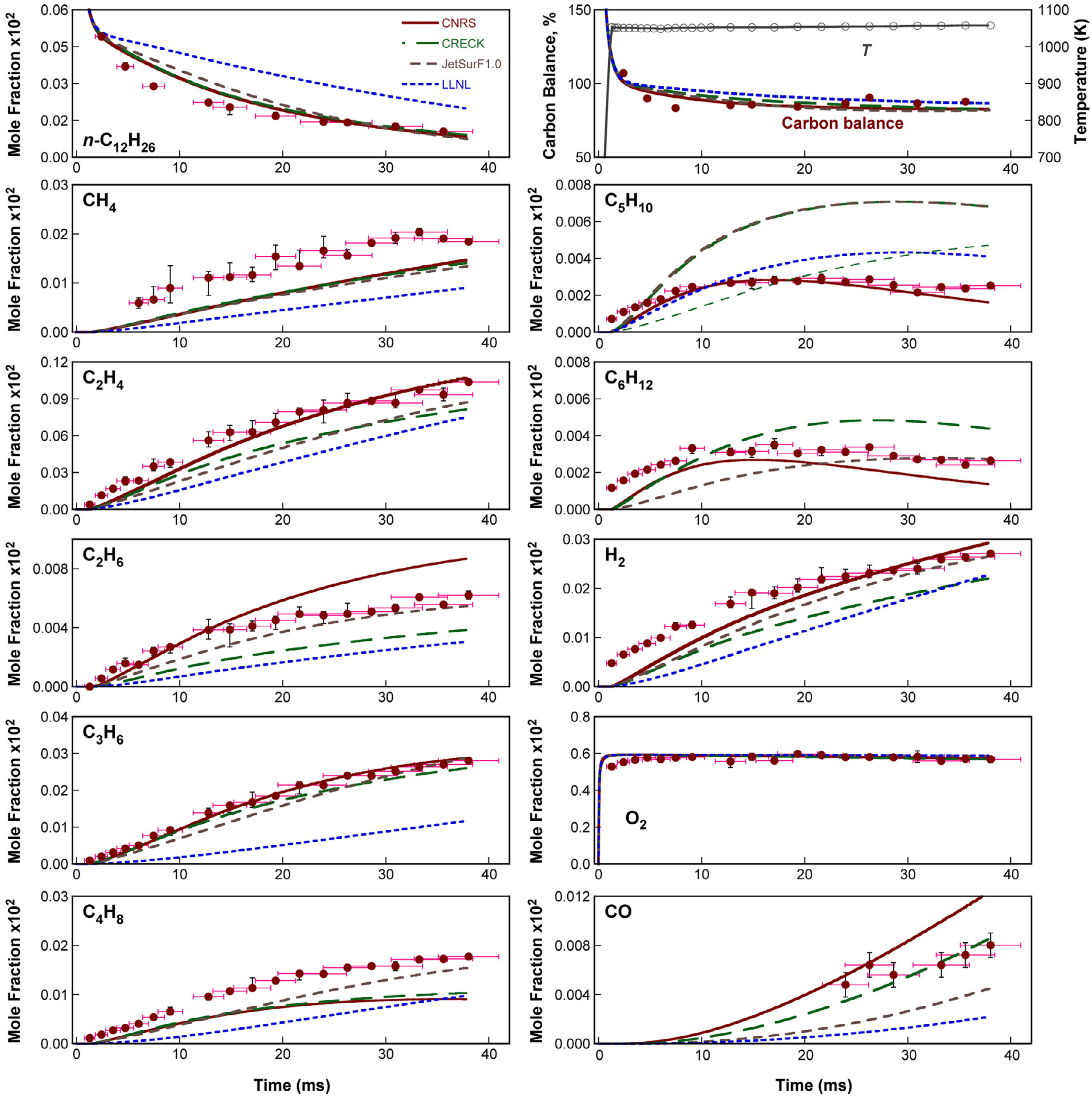

Fig. 4. Species mole fraction, carbon balance and temperature data (symbols) for the 1050 K rich oxidation experiment compared against predictions (lines) from the detailed kinetic models.

initial value to the reactor exit and only a small amount (80 ppmv) of CO generation that amounts to a mere 1.3% of the total fuel carbon during the entire residence time of the experiment. Thus, pyrolytic reactions predominate in the reaction zone and C_2H_4, C_3H_6, 1-C_4H_8 and CH_4 remain the major products of interest. The higher alkenes such as 1-pentene and 1-hexene attain constant concentration values by 10 ms before starting to decrease slightly near the end of the reaction zone. The CNRS model again performs best among the kinetic models tested. The *n*-dodecane decay rate and the production of H_2, C_2H_4, C_3H_6, C_5H_{10} and C_6H_{12} are well predicted. However the model under predicts the rate of CH_4 and 1-C_4H_8 formation as with all other models. JetSurF 1.0 performs fairly well while the LLNL model is the slowest among the four. The experimental carbon balance is between 85 and 90% of the total fuel load. Simulations show that the remaining carbon is present in the higher C_{7-10} alkenes that exist at concentrations of around 5-15 ppmv levels and have not been measured. Overall, all four models perform worse at the 1050 K rich oxidation condition than the 1000 K pyrolysis condition.

The data for the rich 1220 K oxidation experiments are presented in Fig. 5. The equivalence ratio is 1.325 and the injected *n*-dodecane is 569 ppmv. The experimental carbon balance is between 95 and 100% of the fuel load demonstrating that all important carbon containing species have been accounted for. The

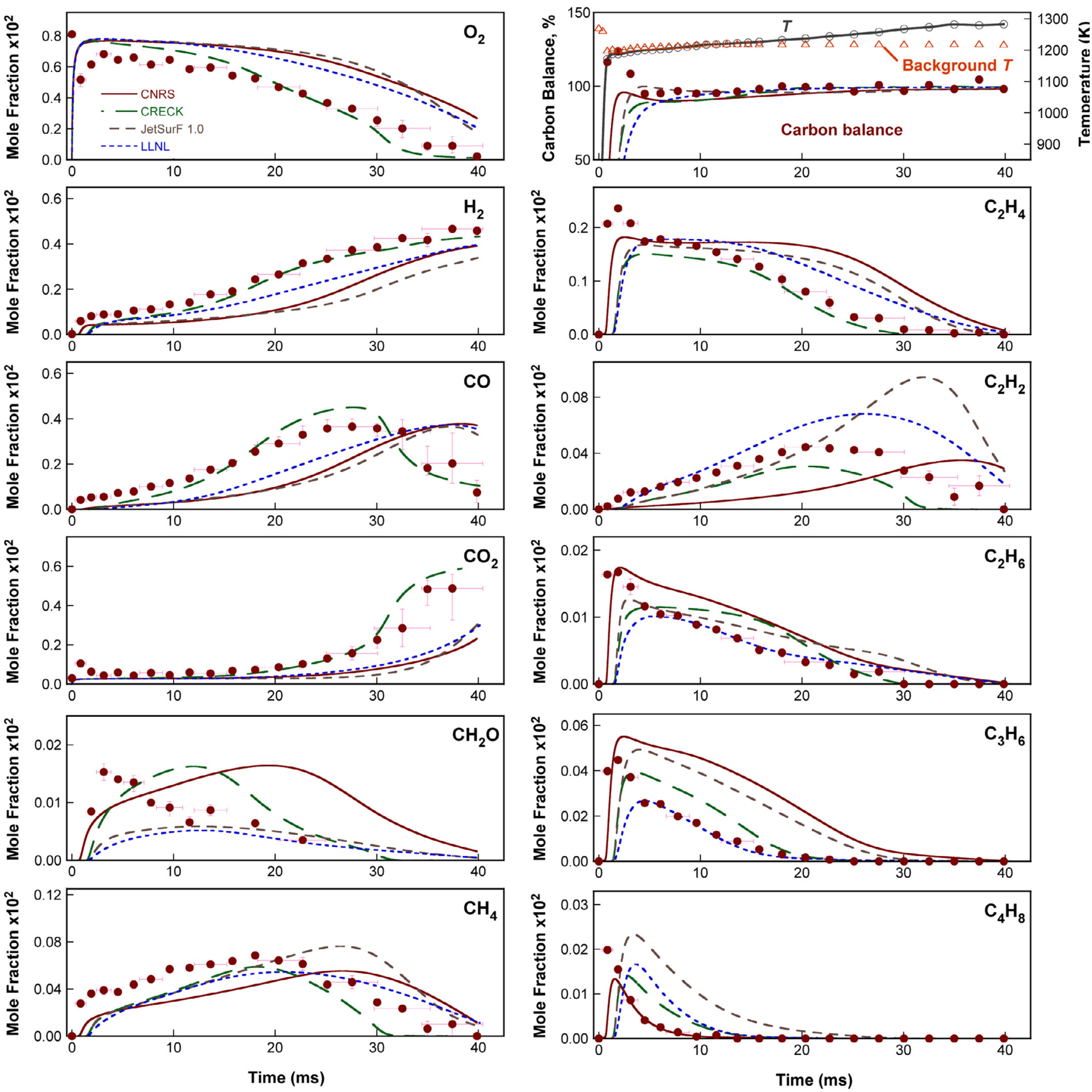

Fig. 5. Species mole fraction, carbon balance and temperature data (symbols) for the 1220 K rich oxidation experiment (symbols) compared against predictions (lines) from the detailed kinetic models.

background flow-reactor temperature under non-reacting conditions is 1220 K. However, under reacting conditions the axial temperature begins to rise after about 16 ms residence time as heat release reactions (primarily the conversion of CO to CO_2) assume importance. The temperature increases to 1280 K near the end of the observed residence time. At these relatively high temperatures, *n*-dodecane decomposes in less than 1 ms and cannot be detected in the flow reactor. Instead the primary hydrocarbon intermediates (1-C_4H_8, C_3H_6, C_2H_4, C_2H_6, C_2H_2 and CH_4), the species containing oxygen (O_2, CO, CO_2 and CH_2O) and H_2 are measured experimentally. The C_{2-4} alkenes, the primary products of *n*-dodecane pyrolysis, peak within the mixing region at 2 ms residence time. The decay of propene and ethylene and the formation of methane and acetylene occur between 4 and 20 ms. Carbon monoxide begins to form at around 15 ms and peaks at 28 ms, after which the conversion of CO to CO_2 dominates the reaction zone. Very little oxygen is consumed in the first 10 ms, but the oxygen concentration drops sharply between 15 and 40 ms when CO and CO_2 formation becomes important. Thus the experimental conditions essentially capture the oxidation of the decomposed alkene fragments such as methane, ethylene, propene and 1-butene to CO and CO_2. JetSurF 1.0, LLNL and CNRS models significantly underpredict the rate of oxidation of the fuel. Thus, for example, the CO peak is predicted to occur at 38 ms while experimentally it is observed at 28 ms residence

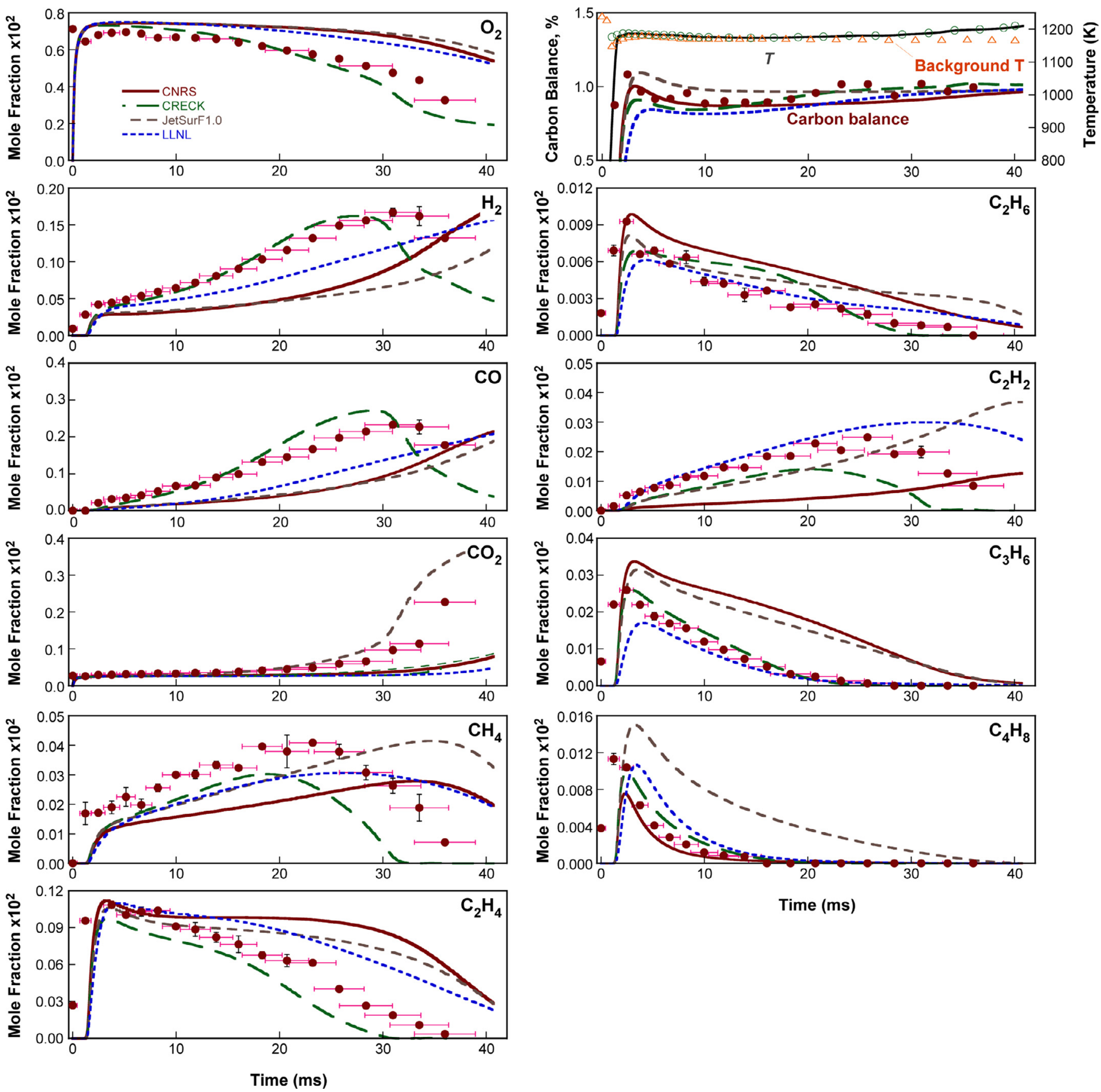

Fig. 6. Species mole fraction, carbon balance and temperature data (symbols) for the 1170 K lean oxidation experiment compared against predictions (lines) from the detailed kinetic models.

time. Similarly the calculated rate of consumption of O_2 and C_2H_4 are far slower than observed. Only the CRECK model is able to predict the CO peak and the decay profiles for C_2H_4 and O_2. Thus the CRECK model predictions for the major species match quite well with the measured data. The agreement worsens for the minor species profiles.

The lean oxidation experiment at 1170 K shows similar trends. The data are presented in Fig. 6. The equivalence ratio is 0.79 and the injected *n*-dodecane is 328 ppmv. The reaction zone temperature begins to rise above the background values from 20 ms into the reaction process. The final temperature is 40 K above the background value of 1170 K by the reactor exit. The experimental carbon balance is 95–100% of the fuel load indicating that all significant carbon containing species have been accounted for. The fuel decomposes rapidly to form C_{2-4} alkenes and the flow reactor captures the reaction zone where these alkenes oxidize eventually to CO and CO_2. Methane and acetylene are byproducts of the oxidation process, and their concentrations peak at around 22 ms, earlier than the CO peak at 32 ms. The H_2 concentration also peaks simultaneously with CO. In a trend similar to the 1220 K rich oxidation experiment, JetSurF 1.0, LLNL and CNRS models predict slower rates of oxidation while the CRECK model predicts the experimental species profiles more closely.

From the experimental results, it is clear that in the temperature regime studied, *n*-dodecane initially decomposes into small alkene fragments (primarily propene and ethylene) through pyrolytic chemistry in which oxygen appears to play little or no part. Around, or above 1150 K however, these alkene fragments react with oxygen to eventually form CO, CO_2 and H_2O. Over the entire range of experimental conditions, none of the models perform a satisfactory job of predicting the oxidation and pyrolysis behavior of *n*-dodecane. For the low temperature pyrolysis and oxidation conditions that capture the initial stages of the *n*-dodecane breakdown, the CNRS and JetSurF 1.0 models make good predictions but the LLNL and CRECK models underestimate the reaction rates. For the higher temperature oxidation conditions that capture the subsequent oxidation of the alkene fragments to CO, CO_2 and H_2O, only the CRECK model is able to reproduce the major species profiles.

4. Optimizing JetSurF 1.0

JetSurF 1.0 is an unoptimized model incorporating evaluated rate parameters ranging from literature compilation to reaction rate rules. With the exception of the H_2/CO submodel that was optimized in the work of Davis et al. [49], no arbitrary parameter adjustments were made in its development. However, each rate parameter has its own uncertainty. In general, a model comprised of thousands of uncertain parameters has little to no possibility to make quantitative predictions for a set of data a priori. The discussion about the comparison of experimental data and model predictions in the previous section is a clear illustration of the above point. Inevitably, large chemical kinetic models for liquid hydrocarbon fuels have to use the principle of chemical similarity to estimate unknown reaction rate parameters. For JetSurF 1.0, the class-rule approach reproduces available experimental observations of *n*-heptane combustion [27] and the laminar flame speed of *n*-dodecane [43] satisfactorily, but it does not provide accurate predictions for the shock tube data available for *n*-dodecane. For example, JetSurF 1.0 predictions show significant differences from multi-species time histories measured for *n*-dodecane oxidation in a shock tube [22]. JetSurF 1.0 predicts a smaller oxidation rate and longer ignition delay for *n*-dodecane than the shock tube measurements—a trend entirely consistent with the observation made for the current flow-reactor experiments. Thus, in at least certain cases, the chemical similarity principle is inadequate. It will require optimization of individual rate parameters against an experimental target data set for a particular fuel in order to improve the overall model accuracy for that fuel.

Recently, Sheen et al. [25–27] proposed a systematic methodology for quantifying and minimizing the effects of kinetic rate parameter uncertainties on the accuracy and precision of the overall model predictions by utilizing well-evaluated experimental data sets as targets. The method is known as the MUM-PCE and combines solution mapping [44] with spectral uncertainty propagation techniques [45]. It can be used in conjunction with a set of experimental targets to advance an optimized model that has an improved nominal prediction and a substantially reduced uncertainty band for its predictions when compared to the trial model. MUM-PCE has been tested on USC-Mech II to develop an optimized model that generates excellent predictions for ethylene combustion [25], along with five to eight times uncertainty reduction in flame speed predictions, a two times decrease of the uncertainty band in ignition delay ($\ln\tau$) predictions and a decrease in the uncertainty of OH concentration predictions with respect to residence time in PSR combustion from a factor of four to a factor of two. The method was also used on JetSurF to arrive at an optimized model for *n*-heptane combustion [27] with the magnitude of accuracy improvements similar to that of the ethylene case.

Multispecies time-history data [22,46] in shock tubes have been shown to impose quite strong constraints on coupled uncertainty of kinetic rate parameters [25]. They provide an efficient and accurate basis for model optimization. Here, we use the *n*-$C_{12}H_{26}$, OH, C_2H_4, CO_2 and H_2O time history data after shock heating of a near-stoichiometric *n*-dodecane/oxygen/argon mixture at an initial temperature of 1410 K [22] as the primary optimization targets. Additional target data considered are atmospheric *n*-dodecane/air laminar flame speed data [24] and ignition delay data for *n*-dodecane/air mixtures [19]. Table 2 summarizes the targets, their values and 2σ uncertainty. Rather than using the current flow reactor data as the optimization targets, we opt to utilize the data to show that when taken together, the multispecies data and the global combustion properties of Table 2 can yield an optimized model that gives an improved prediction quality for experiments under conditions different from those of the target set.

For the species time history data, the experimental targets are either the species concentration at a specific time or the time when the species reaches a specific concentration. The absolute uncertainty of the mole fraction value quoted in [22] is around ±10%, which does not include the impact of temperature uncertainty. In general a 10 K temperature variation leads to around 20% variation in concentration at a given time. Therefore, a 20% uncertainty band for species time history data was adopted. The 2σ uncertainty for the laminar flame speed data is ±2 cm/s, while the uncertainty in experimental ignition delay data was estimated to be ±20% based on regression analysis. Ignition delay is computationally determined from the maximum [OH^*] gradient with time. For this purpose, a set of [OH^*] chemiluminescence reactions [47] are added to JetSurF. The Sandia Premix code [48] is used for flame simulations. Before optimization, the H_2/CO subset rates of JetSurF 1.0 were returned to their as-compiled values [49]. Uncertainty factors for all rate parameters are given previously [27]. Multivariate polynomial response surface are constructed for targets of Table 2. The polynomials are obtained from sensitivity-analysis based method [50] for laminar flame speed and species profiles and central-composite fractional factorial design of resolution VI [51] for ignition delay.

We compare first the simulations from unoptimized JetSurF against the experimental species time history data. In Fig. 7 the dashed lines are experimental data while the solid lines present the model prediction. The experimental uncertainties are given by the error bars on the blackened symbols while model uncertainty scatter (calculated using Monte Carlo sampling) is represented by the colored dots. Firstly, the model uncertainty far exceeds the experimental uncertainty. Secondly, the unoptimized JetSurF predicts a longer induction time as seen by slower OH concentration rise and a delayed disappearance of C_2H_4. Consistent with this trend, the ignition delay times are over-predicted by the model as observed in Fig. 9(a). The uncertainty bands for laminar flames speeds are particularly large as is seen in Fig. 10(a).

Model optimization systematically adjusted the rate parameter values of about two dozen reactions critical to the modeled responses. All of the adjustments were well within the uncertainty of the respective rate expressions. For example, the ratios of the optimized to unoptimized rates are 0.99, 1.24, 0.83, 0.98, 0.42, 1.14 and 1.02 for reactions

$$H + O_2 \rightarrow O + OH$$

$$OH + H_2 \rightarrow H + H_2O$$

$$H + O_2\ (+M) \rightarrow HO_2\ (+M)$$

$$CO + OH \rightarrow CO_2 + H$$

$$HCO + H_2O \rightarrow CO + H + H_2O$$

$$HCO + H \rightarrow CO + H_2$$

$$HCO + O_2 \rightarrow CO + HO_2$$

respectively, in the H_2/CO/small hydrocarbon subset of the model. For reactions involving *n*-dodecane, the ratios are 1.2, 1.3, 1.1, 1.5, and 1.0

Table 2
Experimental data considered and corresponding optimization targets.

Series 1: Shock tube species profile (457 ppm n-$C_{12}H_{26}$/7500 ppm O_2/Ar, $T_5 = \sim 1410$ K, $p_5 = \sim 2.15$ atm)[a] [22]

Species	Target data	Experimental target (uncertainty factor)	Nominal value of prediction
n-$C_{12}H_{26}$	t @ [n-$C_{12}H_{26}$] = 10^{-5}	2.60×10^{-5} (1.25)	2.71×10^{-5} s
C_2H_4	[C_2H_4] @ $t = 10^{-4}$ sec	1.27×10^{-3} (1.20)	1.39×10^{-3}
	t @ [C_2H_4] = 3.49×10^{-4}	6.77×10^{-4} (1.20)	1.16×10^{-3} s
OH	[OH] @ $t = 10^{-4}$ sec	3.47×10^{-6} (1.20)	2.46×10^{-6}
	t @ [OH] = 10^{-5}	4.81×10^{-4} (1.25)	8.54×10^{-4} s
	t @ [OH] = 6×10^{-5}	7.18×10^{-4} (1.20)	1.09×10^{-3} s
	[OH] @ $t = 10^{-3}$ sec	2.57×10^{-4} (1.20)	2.96×10^{-5}
CO_2	t @ [CO_2] = 2×10^{-4}	1.21×10^{-3} (1.20)	1.30×10^{-3} s
H_2O	[H_2O] @ $t = 10^{-4}$ sec	1.58×10^{-4} (1.20)	2.12×10^{-4}
	t @ [H_2O] = 10^{-3}	5.06×10^{-4} (1.20)	7.71×10^{-4} s
	[H_2O] @ $t = 8 \times 10^{-4}$ sec	3.74×10^{-3} (1.20)	1.10×10^{-3}

Series 2: Laminar flame speed (n-$C_{12}H_{26}$/air at $p = 1$ atm and $T_u = 403$ K) [24]

Φ	Experimental target[b] (cm/s)	Nominal value of prediction (cm/s)
0.72	40 ± 2	40.5
0.90	58 ± 2	56.5
1.04	63 ± 2	60.9
1.19	59 ± 2	56.6
1.39	38 ± 2	34.8

Series 3: Ignition delay time (n-$C_{12}H_{26}$/air) [19]

Φ	T_5 (K)	p_5 (atm)	Experimental target (ms) (uncertainty factor)	Nominal value of prediction (ms)
0.56	1117	30.9	266 (1.2)	594
	910	22.7	2134 (1.2)	2200
1.12	1109	23.0	278 (1.2)	360
	907	23.5	1081 (1.2)	641

[a] Actual T_5 and p_5 values: 1392 K and 2.26 atm (n-$C_{12}H_{26}$), 1403 K and 2.41 atm (C_2H_4), 1418 K and 2.35 atm (H_2O), 1404 K and 2.27 atm (CO_2), 1407 K and 2.28 atm (OH).
[b] The $\pm$ values are 2σ uncertainty.

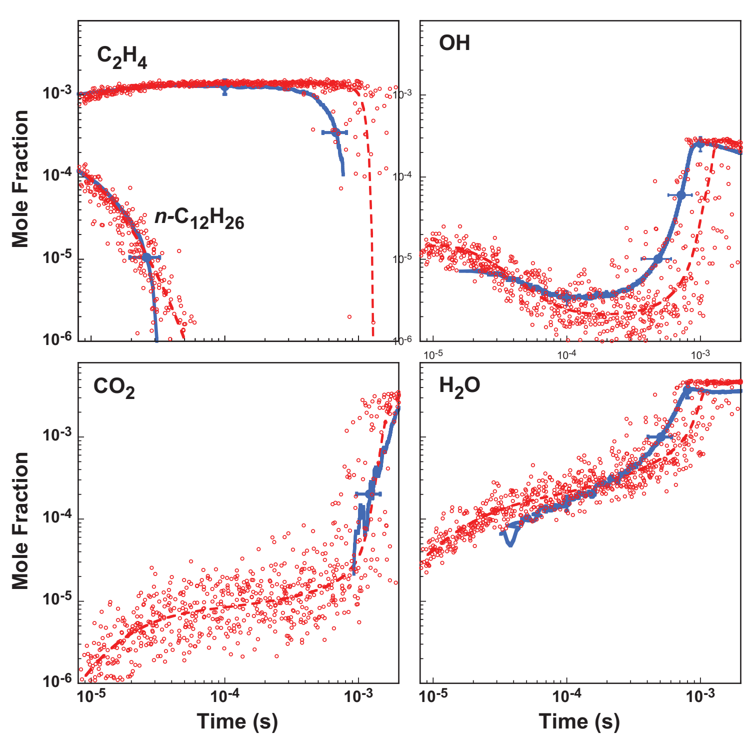

Fig. 7. Experimental (Davidson et al. [22], solid lines) and predicted (unoptimized JetSurF 1.0, dashed lines) species time histories during n-dodecane oxidation in a shock tube (475 ppm n-$C_{12}H_{26}$-7500 ppm O_2 in argon, $T_5 = 1410$ K, $p_5 = 2.15$ atm). Filled circles and error bars: optimization targets and their 2σ uncertainties. Open circles: Monte Carlo sampling of JetSurF 1.0 model uncertainties.

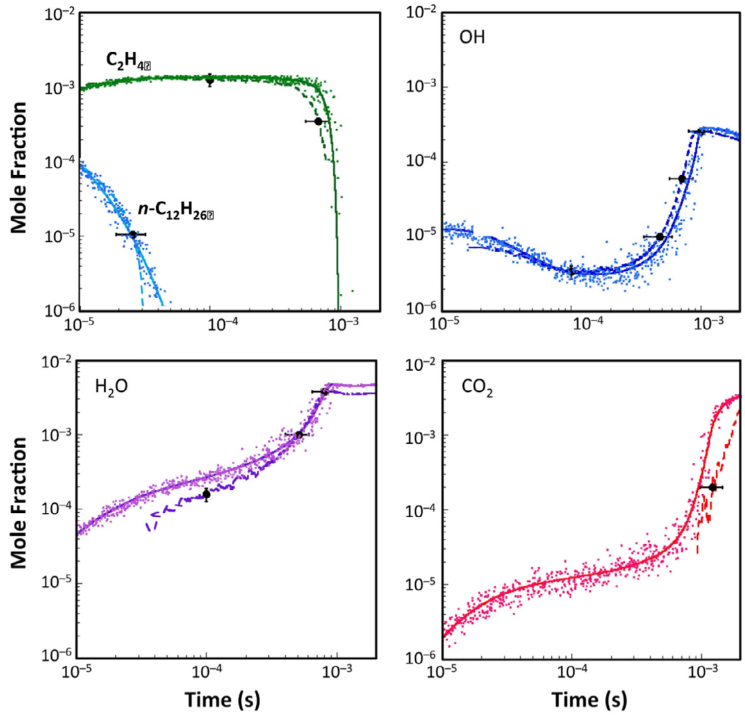

Fig. 8. Experimental (Davidson et al. [22], dashed lines) and predicted (optimized JetSurF 1.0, solid lines) species time histories during n-dodecane oxidation in a shock tube (475 ppm n-$C_{12}H_{26}$-7500 ppm O_2 in argon, $T_5 = 1410$ K, $p_5 = 2.15$ atm). Filled circles and error bars: optimization targets and their 2σ uncertainties. Dots: Monte Carlo sampling of optimized JetSurF 1.0 model uncertainties.

for H abstraction of n-dodecane by the H atom producing the secondary n-alkyl radicals in 2, 3, 4, 5, and 6 positions, respectively

The optimized model is observed to give significantly better predictions and reduced uncertainties for multi-species time histories (Fig. 8), ignition delay time (Fig. 9b) and laminar flame speed (Fig. 10b). The ignition delay times are still somewhat longer than the experimental values; however, the discrepancies are fairly minor. The laminar flame speed predictions have improved considerably as well. The results indicate that once systematic optimization and uncertainty minimization is performed, detailed kinetic models like JetSurF are significantly more quantitative in terms of reproducing experimental data.

The optimized model is tested against n-dodecane data sets that were not used as targets. Malewicki and Brezinsky [23] measured species concentration profiles for n-dodecane pyrolysis and oxidation at elevated pressures (50 atm) in a heated high-pressure single-pulse shock tube. The measurement span the range from 900 to 1700 K in the initial temperature, with residence times ranging from 1.5 to 3.5 ms. The data sets provide a good test for the optimized JetSurF. In Fig. 11, the experimental data are compared with optimized JetSurF predictions using the experimental conditions specified for each shock condition. As can be seen, optimized JetSurF performs well in predicting most of the major species profiles. The model tends to over predict the higher alkene concentration, suggesting that minor pathways to the destruction of these species may be missing. It is noted that Malewicki and Brezinsky [23] also proposed a reaction model for n-dodecane oxidation. Their model reproduces the data shown in Fig. 11 well.

Improvements of model accuracy by optimization may be illustrated by comparing the optimized model against the current flow reactor data, shown in Figs. 12 and 13 for the 1220 K rich oxidation and 1170 K lean oxidation experiments, respectively. The predictions of the unoptimized model are also included in the figures. The match between the optimized JetSurF and the experimental data is far better than the unoptimized model despite the fact that the optimization was carried out using targets under different conditions. For example, for the 1220 K rich oxidation experiment, the optimized model predicts the CO peak to occur at 26 ms while in the experiment it occurs at 28 ms. In contrast, the unoptimized JetSurF predicts the CO peak to occur at 38 ms. The improvements are less significant for the minor products. Similar conclusions can be drawn from the lean oxidation case, though in this instance the overestimation of the rate of oxidation is more significant.

5. Comparison with literature data sets

The current work investigates a relatively narrow but crucial temperature range of the n-dodecane combustion chemistry regime. It is necessary to investigate how well the experimental data set bridges the zone between the low temperature and NTC regime spanning 500–800 K and the high temperature regime above 1300 K.

The decomposition of n-dodecane occurs through multiple pathways. Initiation occurs through unimolecular decomposition, but at the temperature range of interest, H abstraction reactions facilitated by H and OH radicals soon dominate the process. However, it is often convenient to model the decomposition in terms of pseudo-first-order kinetics of the form:

$$n - C_{12}H_{26} \xrightarrow{k_{app}} \text{products}$$

where a pseudo-first-order equation for the fuel mole fraction can be written as

$$\frac{[n - C_{12}H_{26}]}{[n - C_{12}H_{26}]_0} = \exp\left(-k_{app}t\right)$$

and k_{app} is the apparent first order rate constant.

The fuel time-history profile is used to determine k_{app}. MacDonald et al. [52] determined $k_{app}(s^{-1}) = 7.46 \times 10^{12} \exp(-27300/T)$ from n-dodecane decomposition for temperatures ranging from 1600 to 1100 K from several sources [52,53]. However, there are very few measurements in the intermediate 900–1100 K region, and the current calculations prove useful to bridge the gap. The flow reactor

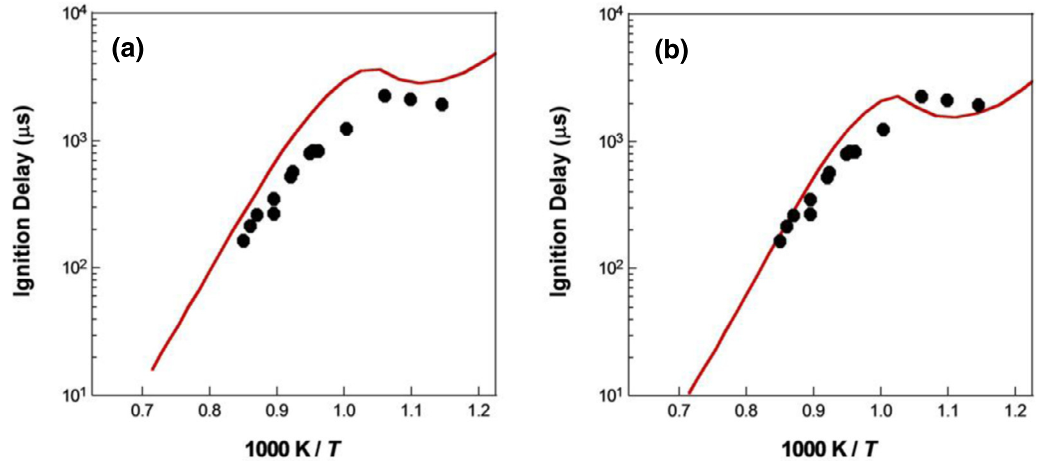

Fig. 9. Experimental ([19], symbols) and predicted (lines) ignition delay times. (a) Unoptimized JetSurF 1.0; (b) optimized JetSurF 0.565% n-$C_{12}H_{26}$/20.89%O_2/N_2 and $p_5 = 20$ atm.

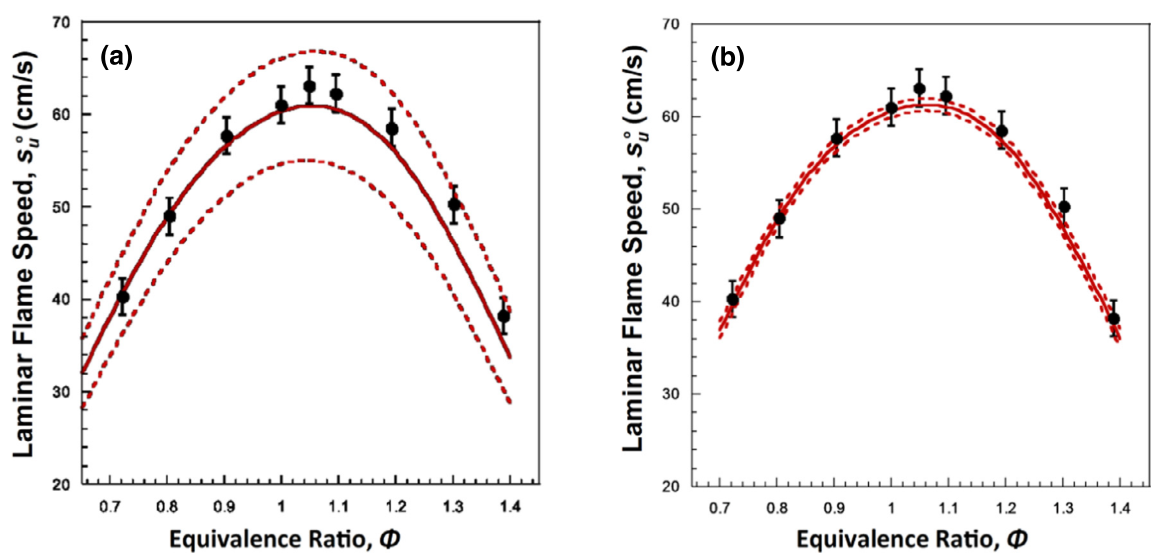

Fig. 10. Experimental ([24], symbols) and predicted (lines) laminar flames speeds of n-dodecane-air mixture at 1 atm pressure and an unburned gas temperature of 403 K. (a) Unoptimized JetSurF 1.0; (b) optimized JetSurF. The dashed lines indicate the 2σ uncertainties of the model.

operates under vitiated conditions with 22% water present in the bath gas, the rest being primarily nitrogen. As a third body, water molecules are more efficient than inert molecules like nitrogen or argon and accelerate the rate of fuel decomposition. Based on the water concentration, the experimentally determined decomposition rate values are divided by a correction factor of 2.1. This value was obtained through modeling comparing the rate of n-dodecane disappearance with and without the amount of water added. Figure 14 shows that the calculated value of k_{app} for the 1000 K and 1050 K conditions matches well with rate expression, after correcting for the presence of water in the flow reactor environment.

The peak or the plateau value for ethylene, the most important product of decomposition, is also an important target for kinetic schemes. MacDonald et al. [21] obtained data sets plotting the dependence of peak ethylene yield versus temperature under pyrolytic conditions. The current study extends this to lower temperatures. As seen in Fig. 15, the present data are consistent with the high temperature data. The ethylene yields were measured at 19 atm pressures while the current data are measured at atmospheric pressure. However, the n-dodecane decomposition rate and the ethylene peak values are relatively insensitive to pressure as the pressure-dependent reactions relevant for fuel pyrolysis are mostly near the high-pressure limit. Therefore, no significant discrepancies are expected from the pressure mismatch.

Herbinet et al. [16] studied the thermal decomposition of n-dodecane (at 2% mole with balance helium) in a JSR between 773 and 1073 K and 1 atm pressure. The particular profiles of interest are the fuel and product species profiles at 973 K as the temperature conditions are quite close to the current 1000 K pyrolysis experiment. A comparison of the fuel conversion and the product mole fractions normalized by the fuel load is shown in Fig. 16. The optimized JetSurF model has been employed to perform CHEMKIN simulations for the conditions in the JSR experiments. The plots make it clear that the presence of excess H_2, a source of H radicals, and the vitiated conditions of the current flow reactor experiment substantially accelerate the rate of n-dodecane decomposition in comparison to the inert background gas conditions for the JSR pyrolysis at similar temperature and pressure conditions. The optimized JetSurF model replicates this effect, though it predicts the flow reactor results better than the data obtained from the JSR experiments.

6. Model comparisons and analyses

It has been shown that the rate of n-alkane oxidation at high-temperature is primarily determined by the kinetics of small

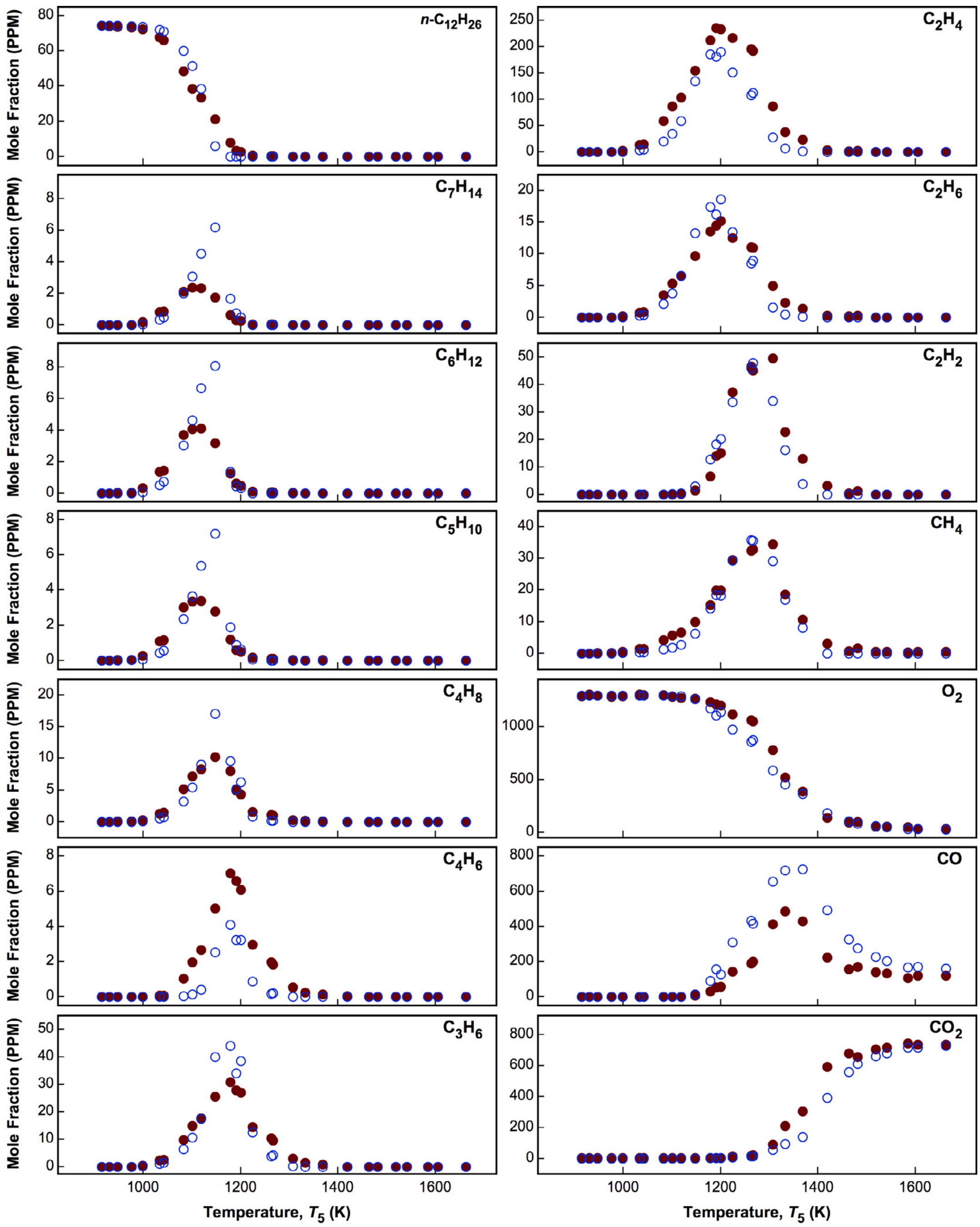

Fig. 11. Comparisons of experimental ([23], filled symbols) and computed (optimized JetSurF model, open symbols) species mole fractions during *n*-dodecane oxidation over a range of temperatures T_5; 75 ppm *n*-dodecane, $\Phi = 1.06$, balance argon, nominal pressure of 50 atm, and dwell time from 1.2 to 2.4 ms. The computation at each T_5 value was made under the corresponding conditions of the particular experiment.

hydrocarbon fragments and H_2/O_2 reactions [24,43]. Thus the differences observed among the kinetic models tested here are likely caused by the differences in the kinetic parameters used for the small hydrocarbon chemistry. Here, sensitivity and flux analyses are carried out for all four models to examine these differences. The optimized JetSurF and the LLNL model are chosen as examples for discussion. The principal conclusions to be reached are independent of the model choices.

The 1000 K pyrolysis and the 1050 K rich oxidation runs sample the initial fuel breakdown stage of the reaction that produce ethylene and propene as the main decomposition products. There is general agreement regarding the basic kinetic pathways through which large *n*-alkanes undergo thermal decomposition [10,13,14,54]. The formation of alkyl radicals from the *n*-alkane occur mainly through H-abstraction reactions although direct C–C fission reactions may dominate at temperatures above 1400 K. At the flow reactor conditions, reaction proceeds through H-abstraction and successive β-scission of the alkyl radicals. The large alkenes undergo either H-abstraction to form alkenyl radicals or H-addition reactions followed by β scission to form smaller products. The basic pathways are similar for the different models tested and a sample flux analysis is shown in Fig. 17 from the optimized JetSurF model for the 1000 K pyrolysis condition at the time point when 150 ppm of *n*-dodecane remains unreacted (approximately 47% fuel decay). The alkyl radicals also undergo mutual isomerization (not shown) while undergoing β-scission. The decomposition paths for the alkenes are also omitted.

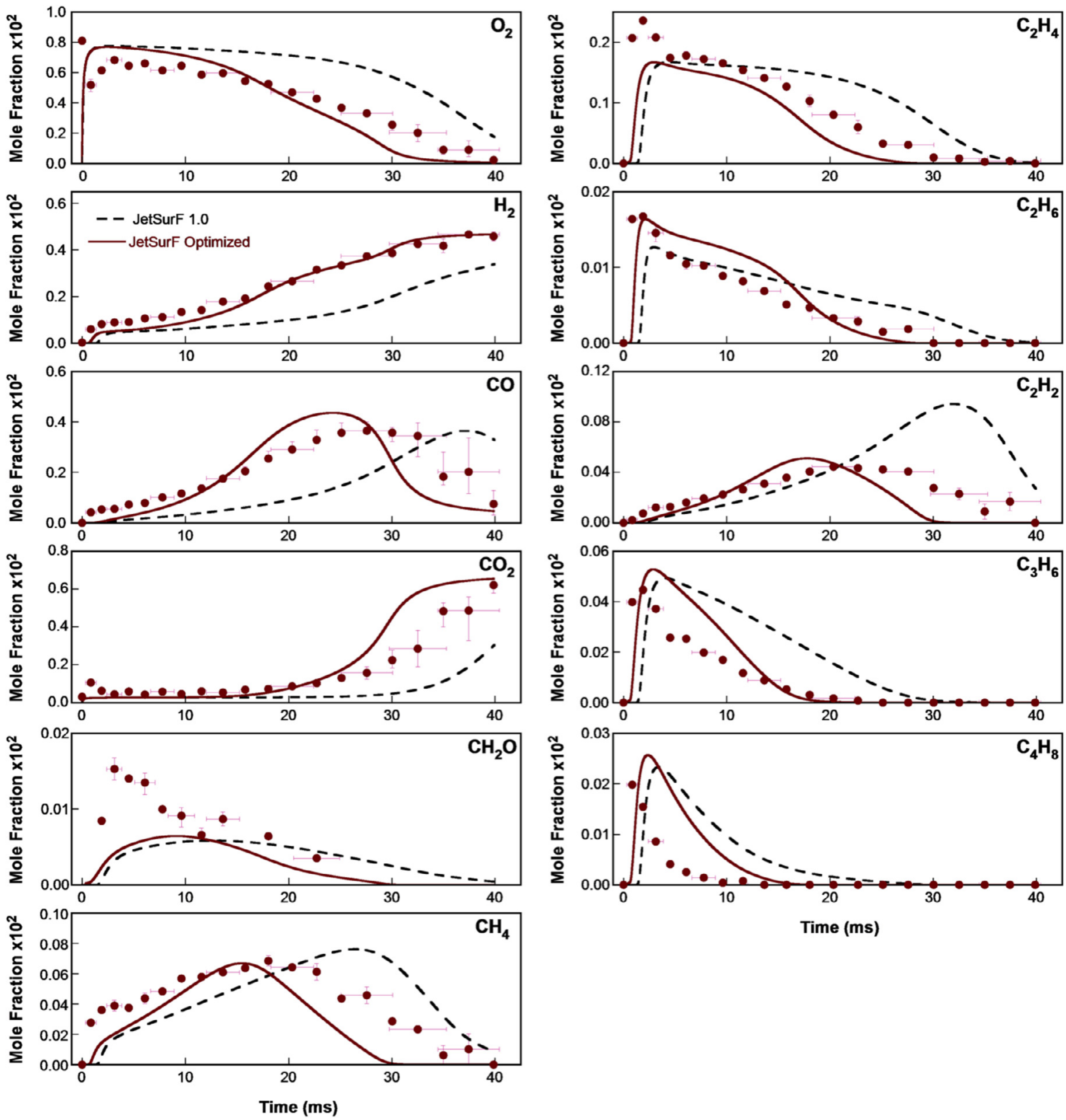

Fig. 12. Comparisons of the species mole fraction profiles in the 1220 K rich oxidation experiment (symbols) against predictions of optimized and unoptimized JetSurF 1.0 (lines).

As can be seen, the dodecyl isomers dissociate to create smaller alkyl fragments which in turn undergo a new round of β scission to eventually generate butyl, propyl and ethyl radicals. These radicals are the dominant sources of ethylene, the primary pyrolysis product. A rate of progress analysis shows that for the 1000 K pyrolysis conditions, the H-abstraction reactions proceed primarily through H-atom attack. For the 1050 K rich oxidation conditions, the pathways are nearly identical, however H-abstraction by OH also plays a role. For the 1000 K pyrolysis condition, the sensitivity analysis was done at the residence time corresponding to the 47% fuel decay point. The ranked logarithmic sensitivity coefficients of n-dodecane concentration under pyrolysis conditions are shown in Fig. 18 for the optimized JetSurF and the LLNL models. In the figure, reactions with positive sensitivities are those that retard fuel decomposition while reactions with negative sensitivities are those that enhance fuel decomposition. Unsurprisingly, the H abstraction reaction rates show negative sensitivity as they are primarily responsible for the fuel decay. However, the rate at which H abstraction reaction can take place depends on the concentration of the H atom. Thus, the availability of the H radical determines the rate of n-dodecane decomposition. For both models, two reactions are responsible for producing almost all of the H radicals. They are,

$$C_2H_5(+M) \rightarrow C_2H_4 + H(+M) \tag{R1}$$

$$CH_3 + H_2 \rightarrow CH_4 + H \tag{R2}$$

For the optimized JetSurF, these reactions run faster than the LLNL model such that the production rate of H radicals are 2.5-1.5 times that in the LLNL model during the initial 10 ms of the residence time. Reaction R2 is one of the primary sources of H radicals under pyrolysis conditions where a large excess of H_2 (6.6%) is present in the background gas driving the reaction in the direction of CH_4 formation. The dominance of Reaction R2 also explains why under the current flow reactor conditions of excess hydrogen, pyrolysis proceeds faster than what is generally observed in a pure inert gas dilution.

Reaction R2 remains important for the LLNL model as well. However, the distinctive feature of the LLNL model is the increased

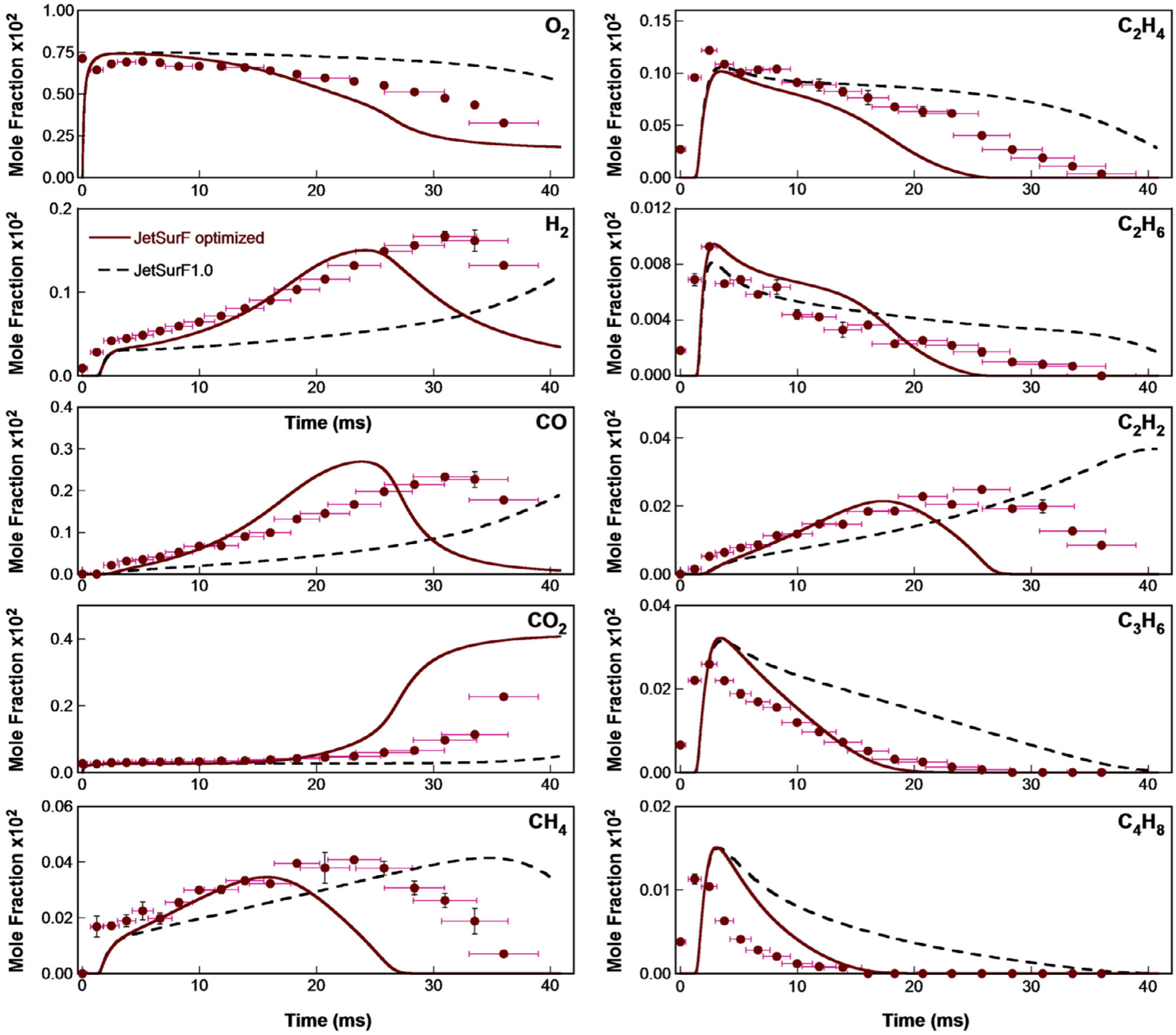

Fig. 13. Comparisons of the species mole fraction profiles in the 1170 K lean oxidation experiment (symbols) against predictions of optimized and unoptimized JetSurF 1.0 (lines).

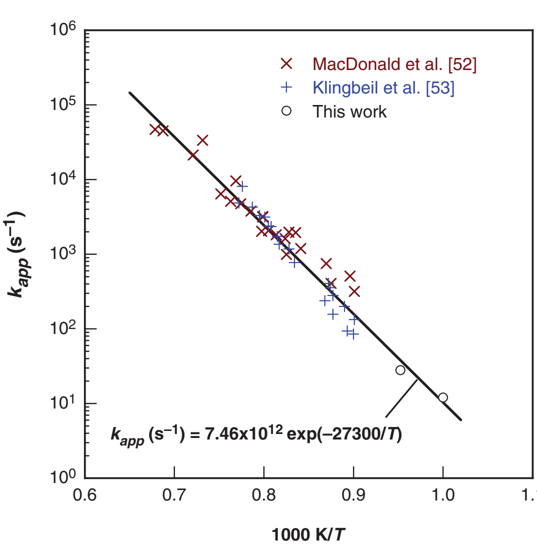

Fig. 14. Apparent first-order rate constant of *n*-dodecane pyrolysis.

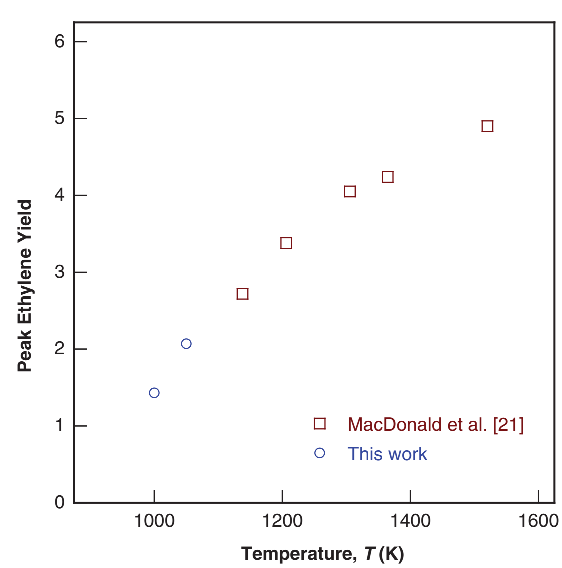

Fig. 15. Peak ethylene yield versus pyrolysis temperature.

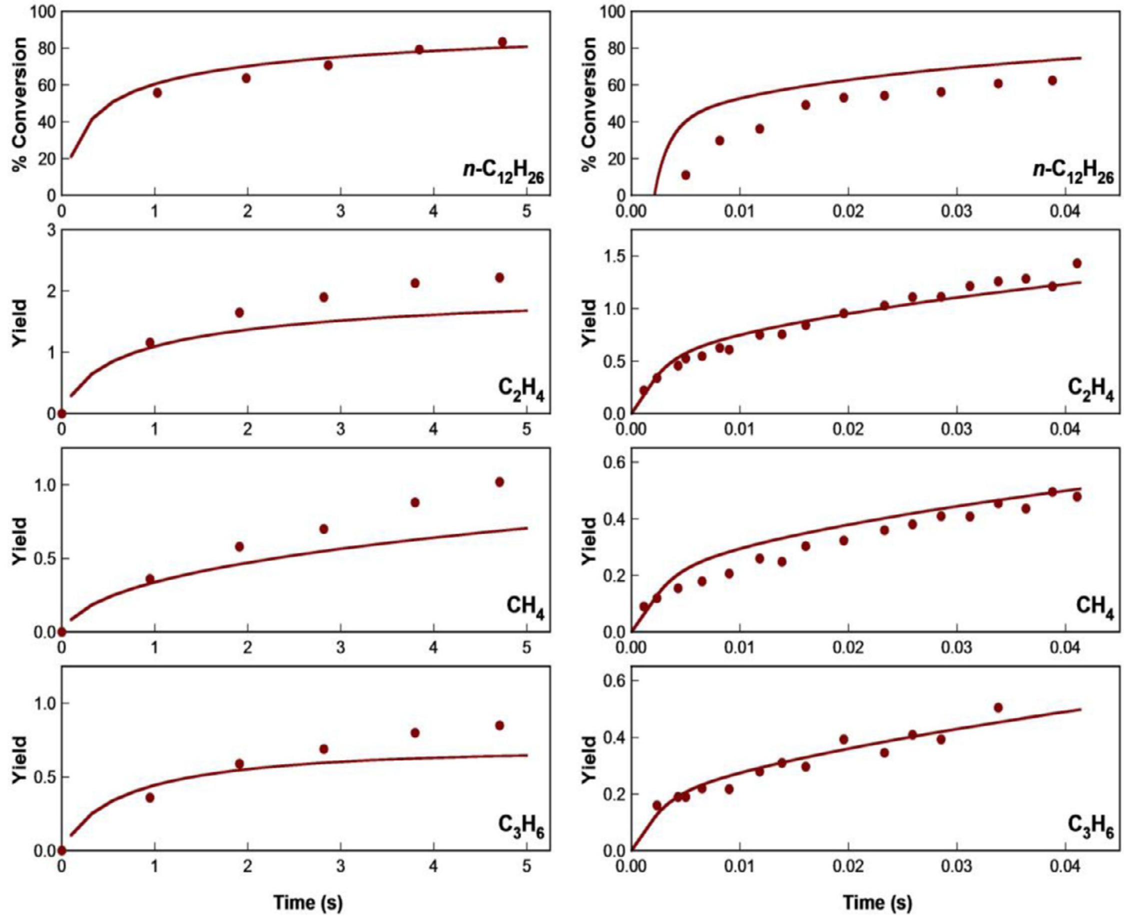

Fig. 16. Fuel conversion and species yield data of the current 1000 K flow-reactor pyrolysis experiment (right panel) compared against the 973 K JSR pyrolysis data of *n*-dodecane [16] (left panel). Lines are predictions of the optimized JetSurF.

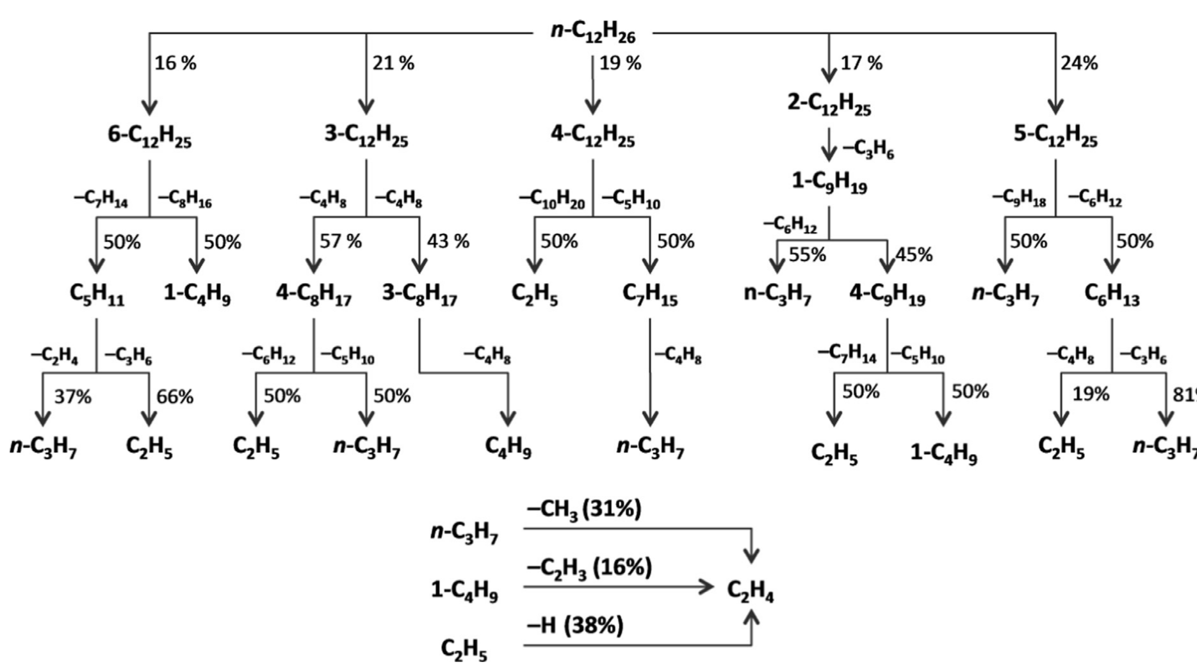

Fig. 17. Major reaction flux analysis for the 1000 K *n*-dodecane pyrolysis conditions as predicted by the optimized JetSurF model when 150 ppm of *n*-dodecane remains unreacted.

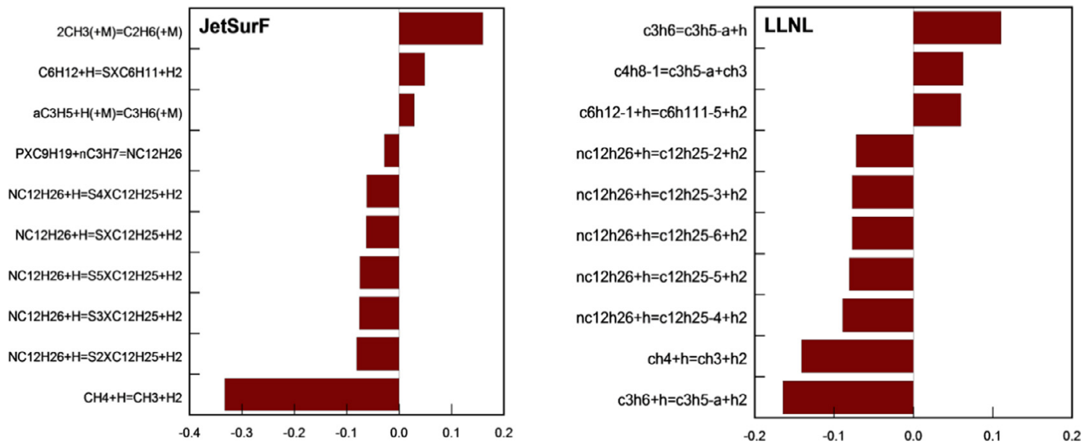

Fig. 18. Ranked logarithmic sensitivity coefficients of *n*-dodecane concentration when 150 ppm of the reactant remains under the 1000 K pyrolysis conditions, comparing the optimized JetSurF and LLNL models.

Table 3
Reaction rate coefficient (units: mol, s, cc) comparisons at 1000 K between the optimized JetSurF and the LLNL models for the most influential reactions pertaining to the 1000 K pyrolysis and 1050 K rich oxidation experiments.

Reaction	k (JetSurF)	k (LLNL)	Ratio	Effect
n-$C_{12}H_{26}$ +H → $C_{12}H_{25}$ +H_2 (avg.)	5.36×10^{12}	4.34×10^{12}	1.235	Fuel decay
n-$C_{12}H_{26}$ +OH → $C_{12}H_{25}$ +H_2O (avg.)	6.46×10^{12}	6.47×10^{12}	1	Fuel decay (oxidation)
CH_3 +H_2 → CH_4 +H	1.18×10^{10}	0.81×10^{10}	1.45	Enhances pyrolysis
CH_3 +H_2O→ CH_4 + OH	2.01×10^{8}	1.80×10^{8}	1.1	Enhances oxidation
aC_3H_5 +H_2 → C_3H_6 +H	3.00×10^{7}	2.63×10^{8}	0.11	LLNL runs in this direction
C_3H_6 +H → aC_3H_5 +H_2	1.17×10^{12}	1.56×10^{12}	0.75	JetSurF runs in this direction
C_2H_5 (+M) → C_2H_4 +H (+M)	5.66×10^{4}	2.14×10^{4}	2.6	Main source of ethylene
aC_3H_5 +H (+M)→ C_3H_6 (+M)	1.88×10^{14}	1.15×10^{14}	1.6	Retards pyrolysis

importance of the allyl (aC_3H_5) pathways, particularly the reactions R3 and R4, in determining the fuel decay sensitivities.

$$C_3H_6 + H \rightarrow aC_3H_5 + H_2 \tag{R3}$$

$$aC_3H_5 + H \rightarrow C_3H_6 \tag{R4}$$

Under pyrolysis conditions, the LLNL model has reaction R3 as the largest sink of allyl radicals and a source of H radicals, while in the optimized JetSurF model the reaction proceeds in the reverse direction and acts as one of the sources of allyl radical and a minor sink for H radicals. This switching of the direction of reaction progress can be explained from the rate coefficient values presented in Table 3, where it is seen that the reaction coefficient of the back reaction of R3 in the LLNL model is 10 times larger than that of JetSurF. This discrepancy is primarily due to the different thermochemical data used. JetSurF follows recommendations from the Burcat thermochemical database [55], while the LLNL model uses the group additivity based estimates generated by the THERM software [56] that uses a somewhat older group property database [57]. The Gibbs energy of formation for propene differs by 2.32 kcal/mol and that for the allyl radical differs by 1.4 kcal/mol at 1000 K between the two models. As a result the ΔG^o of reaction R3 for the JetSurF model is –17 kcal/mol while the value for the LLNL model is -21 kcal/mol. Such differences lead to an equilibrium constant (K_c) value that is 6.5 times greater in JetSurF than in the LLNL model.

In the optimized JetSurF model, the large positive sensitivity associated with the methyl recombination indicates that reaction R5 is a major sink of the radicals.

$$2CH_3\,(+M) \rightarrow C_2H_6\,(+M) \tag{R5}$$

In the LLNL model, a similar inhibiting role is played by the combination of methyl and allyl, i.e., reaction R6.

$$aC_3H_5 + CH_3 \rightarrow 1\text{-}C_4H_8 \tag{R6}$$

Therefore, much of the differences between the LLNL and JetSurF fuel decay rates can be attributed to different thermochemical data assigned to allyl and propene.

In the rich oxidation run (1050 K, 1.56 equivalence ratio), water molecules replaces hydrogen as the third body mediating the most dominant methyl to methane conversion.

$$CH_3 + H_2O \rightarrow CH_4 + OH \tag{R7}$$

Reaction R7 is the most influential reaction responsible for enhancing the *n*-dodecane decomposition rate for both JetSurF and LLNL models by acting as a source of OH radicals.

Under the 1220 K rich oxidation run conditions, *n*-dodecane decays rapidly into ethylene and propene within the first millisecond and the primary chemistry that occurs in the flow reactor involves the oxidation of ethylene and propene into C_2H_2, CO and eventually CO_2. The reaction flux diagram at 2500 ppm CO formation point for the optimized JetSurF is shown in Fig. 19. The primary pathway of ethylene oxidation follows the well-known pathways,

$$C_2H_4 + OH \rightarrow C_2H_3 + H_2O \tag{R8}$$

and

$$C_2H_3 + O_2 \rightarrow CH_2CHO + O\,(64\%) \tag{R9}$$

$$C_2H_3 + O_2 \rightarrow HCO + CH_2O\,(10\%) \tag{R10}$$

$$C_2H_3\,(+M) \rightarrow C_2H_2 + H\,(+M)\,(25\%) \tag{R11}$$

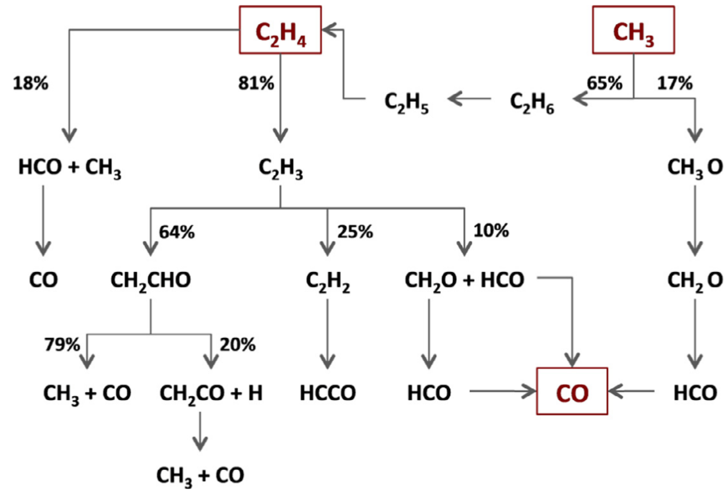

Fig. 19. Reaction flux analysis for the rich oxidation experiment of *n*-dodecane at 1220 K as predicted by the optimized JetSurf model when CO mole fraction reaches 0.25%.

The vinoxy (CH_2CHO) radical rapidly breaks down to CH_3 and CO. Acetylene persists, but also undergoes O atom attack to form HCCO that eventually oxidizes to CO. Formaldehyde undergoes OH attack to generate the formyl (HCO) radical. HCO acts as the primary source of CO in the system. The CH_3 radical also plays an important role. Most of it recombines to form C_2H_6 that in turn generates C_2H_5 radical and is converted back to C_2H_4. The rest react with O atoms to form formaldehyde (CH_2O), or H atoms to form CH_4. The primary source of OH radicals is from the H+O_2 chain branching step (R12), which is by far the most influential reaction affecting ethylene formation rates at these late stage oxidation conditions.

$$H + O_2 \rightarrow O + OH \tag{R12}$$

An examination of the sensitivities and fluxes of ethylene oxidation reveal that the primary difference between the optimized JetSurF and LLNL models lies in the rate coefficients assigned to the $C_2H_3 + O_2$ reactions (R9 and R10). The rate expressions used in both models are perhaps outdated and will require revisions [58]. Regardless, the current analysis again supports the earlier findings (laminar flame speed sensitivities [43] and ignition delay sensitivities [22]) that a predictive model for the oxidation of *n*-dodecane at temperature beyond that of the NTC region relies largely on an accurate model for the foundational fuel (H_2, CO, C_{1-4}) chemistry, as the oxidation rates of the fragments of *n*-dodecane decomposition are rate limiting during *n*-dodecane combustion. Thus, future efforts in kinetic model development should be placed proportionally more on improving the accuracy of the foundational fuel chemistry.

7. Concluding remarks

The present work investigated the kinetics of *n*-dodecane pyrolysis and oxidation. Experimentally, species time profiles were obtained in the intermediate temperature regime of 1000–1250 K under atmospheric pressure in the Stanford VPFR. The work extended the experimental database of *n*-dodecane oxidation over this critical but somewhat neglected temperature region of *n*-alkane combustion. The flow reactor was modified to handle low vapor-pressure hydrocarbon fuels and species time-history profiles were obtained for the *n*-dodecane, oxygen and 12 intermediate and product species over a 1–40 ms residence time using real-time gas chromatography. The repeatability and carbon balance were excellent under all conditions. The experiments were conducted at four initial conditions to investigate fuel pyrolysis as well as early and late stages of oxidation chemistry under both rich and lean conditions. The experimental results were compared against the predictions of four chemical kinetic models available in the literature. It was found that the models can diverge quite notably in their predictions of the concentration profiles of intermediate and product species among themselves and against the data collected under the current flow reactor conditions.

Computationally, JetSurF 1.0 was subject to optimization and uncertainty minimization independently against a selected set of *n*-dodecane combustion target data available in the literature using MUM-PCE. The flow reactor data were used for the purpose of testing the optimized model. The optimized model is shown to predict a wide range of *n*-dodecane pyrolysis and oxidation data at high temperature better than the trial model both in terms of improved accuracy and narrowed uncertainties. The MUM-PCE procedure improved the model predictions against the current flow reactor data notably, thus illustrating the usefulness of the methodology.

In agreement with findings of earlier studies, the overall process of *n*-dodecane oxidation at temperature above 1000 K is found to be separated consistently into two stages: fuel pyrolysis and oxidation of the pyrolysis products, including mainly hydrogen, ethylene, propene, 1-butene and methane. Reactions in the second stage of the process are always rate limiting. Thus, it is not surprising to find that the divergent predictions in the literature reaction models of *n*-dodecane oxidation are primarily caused by the differences in the respective small hydrocarbon chemistry submodels. For this reason, improvements of our ability to model the oxidation kinetics of *n*-dodecane and for that matter, all practical liquid fuels will require a better knowledge of the reaction chemistry of H_2/CO/C_{1-4} hydrocarbons.

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Appendix A

The optimized JetSurF model file in CHEMKIN format. (Available in Supplementary material).

Supplementary materials

Supplementary material associated with this article can be found, in the online version, at doi:10.1016/j.combustflame.2015.08.005.

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